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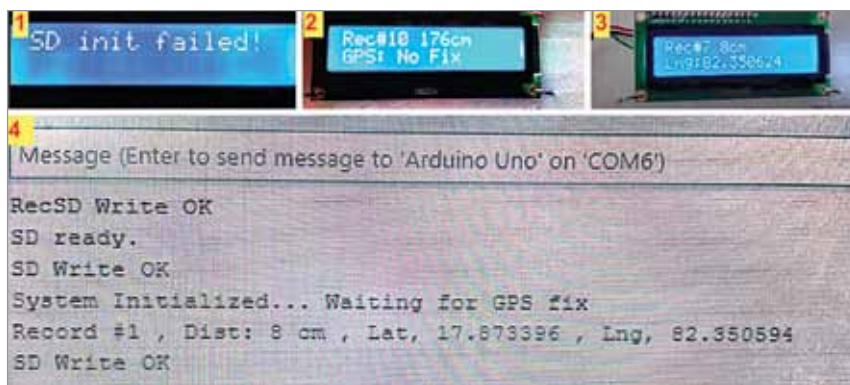


Fig. 6: Device displaying status and serial output of logged data on the SD card

hardware integration.

Connect the 16 × 2 I²C LCD to the Arduino Uno's SDA and SCL pins (A4 and A5) for the real-time display of depth and system status. Interface the HC-SR04 ultrasonic sensor with the designated digital pins (trigger and echo) to enable accurate distance measurement. The GPS module is connected using SoftwareSerial communication, for example, pins D2 and D3, to acquire real-time latitude and longitude coordinates for geotagging.

Next, connect the SD card module via the SPI interface using pins D10 (CS), D11 (MOSI), D12 (MISO), and D13 (SCK) to reliably store depth and location data in CSV format. A push button is integrated to initiate or control the logging process, while a buzzer is connected to a digital pin to provide audible feedback on system status, such as successful data logging or an error indication.

Ensure that all components are powered with appropriate voltage levels and share a common ground to maintain signal integrity and

stable operation. Proper wiring and secure connections are essential to avoid data loss or measurement errors during field use.

Finally, mount the fully assembled system onto a PVC pipe, as shown in Fig. 5. This handheld frame provides mechanical stability, ease of handling, and suitability for field deployment in OFC-laying operations, enabling efficient and accurate trench-depth measurement and location tracking.

Now, power on the device. If the SD card is not detected, the display will show 'No SD Card.' Insert a properly formatted SD card into the slot; the system will then begin displaying depth measurements. Allow a few seconds for the GPS module to acquire satellite signals and obtain a valid fix. Once the GPS achieves a valid fix, the device displays coordinates and begins logging trench depth with location data. The same information can also be observed on the serial monitor, accompanied by a buzzer beep for each recorded entry (see Fig. 6). To view the logged data, remove the SD card, insert it into a card reader, and open the file to access the stored records. **EFY**

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